Marcel Torné Villasevil
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EDUCATION

- **Harvard University** — Cambridge, MA
  Master of Engineering in Computational Science and Engineering; GPA: 4.00/4.00 — *Aug. 2021 – May 2023*
- **Massachusetts Institute of Technology** — Cambridge, MA
  Master of Engineering Thesis and Cross-Registering student; GPA: 5.00/5.00 — *Aug. 2021 – May 2023*
- **École Polytechnique Fédérale de Lausanne, EPFL** — Lausanne, Switzerland
  Bachelor of Science in Computer Science; Grade: 5.62/6 (top < 3%) — *Sept. 2018 -- July. 2021*

CONFERENCE PUBLICATIONS

- ***Breadcrumbs to the Goal: Goal-Conditioned Exploration from Human-in-the-loop feedback***
  Marcel Torné Villasevil, *Max Balsells I Pamies, Zihan Wang, Samedh Desai, Tao Chen, Pulkit Agrawal , Abhishek Gupta*
  Submitted to the Neural Information Processing (NeurIPS) 2023
- ***Autonomous Robotic Reinforcement Learning with Asynchronous Human Feedback***
  Max Balsells I Pamies,* ***Marcel Torné Villasevil****, Zihan Wang*, Samedh Desai, Pulkit Agrawal , Abhishek Gupta*
  Submitted to the Conference on Robot Learning (CoRL) 2023
  * indicates equal contribution

RESEARCH EXPERIENCE

- **Massachusetts Institute of Technology CSAIL** — Cambridge, MA
  Graduate researcher supervised by Prof. Pulkit Agrawal and Prof. Abhishek Gupta — *Feb 2022 – Present*
  - My research is on leveraging Human Feedback to teach new skills to robots in the real world. I am tackling this problems in three different lines, first, designing new algorithms to incorporate noisy human feedback for learning policies, designing new algorithms for reset-free learning in the real world, and working on new ways to close the real to sim to real gap.
- **Microsoft** — Bellevue, WA
  Data Scientist Intern — *June 2022 - August 2022*
  - **Microsoft Bing:** Built the first implementation of Extreme Classification algorithms by Microsoft Research in the Image ranker of Bing.
- **EPFL - Machine Learning and Optimization Lab (MLO)** — Lausanne, Switzerland
  Bachelor's Thesis research supervised by Prof. Martin Jaggi — *Feb 2021 - July 2021*
  - Developed a mobile application in JavaScript for privacy-preserving decentralized Machine Learning in medical image classification. [https://github.com/epfml/disco]
- **EPFL - Formula Student Racing Team**
  Autonomous driving team — *Feb 2021 - July 2021*
  - At the EPFL Racing Team, I pursued an individual Machine Learning Research Project. More concretely, I developed an algorithm for the Control and Planning of an autonomous vehicle using Deep Reinforcement Learning.

SELECTED HONORS

- **Harvard Real Colegio Complutense Scholarship** — *Aug. 2021 - May 2023*

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Merit-based scholarship for Spanish students studying at Harvard. Covers around 70% of the Master's tuition.

- **Harvard IACS Master Thesis proposal award** *Aug. 2021 - May 2023*
  Award given to a small, selected number of students in the master's program at Harvard for their excellent Master's thesis proposal. Covers around 10% of the total master's tuition.

- **Nova 111 Student List 2022 Spain** *May 2022*
  I was selected as one of the top 11 best students from Spain in Computer Science aged between 18-25

- **Excellence in the University Entrance Exam (top < 1%)** *July 2018*
  Award given by the Catalan president himself

- **Bronze Medal Spanish National Physics Olympiad 2018** *June 2018*

- **City of Reus High School Academic Excellence Award** *September 2016*
  Price given to the best student of each high school by the mayor

TEACHING EXPERIENCE

- **École Polytechnique Fédérale de Lausanne, EPFL** Lausanne, Switzerland
  Teaching Assistant in Computer Architecture *Sept. 2020 – Jan. 2021*
  - Lead a team of a dozen highly skilled graduate students. I owned the major project of the course, programming the Game of Life by John Horton Conway in assembly code.

- **École Polytechnique Fédérale de Lausanne, EPFL** Lausanne, Switzerland
  Teaching Assistant in General Physics I: Mechanics *Sept. 2019 - Dec. 2019*
  - Weekly meetings with a group of 20-40 students, I assisted reviewing lectures and helping students out with problem sets.

INDUSTRY EXPERIENCE

- **Amazon** Barcelona, Spain
  Software Engineering Intern *May 2020 - September 2020*
  - **Amazon Search, Science and AI:** *Developed browser plugin tool using Angular JS to help ML engineers debug their ranking algorithms. Tool used by more than 100 times per day by more than 20 engineers.*

LEADERSHIP EXPERIENCE

- **Quiz.Fit** Lausanne, Switzerland
  Co-founder and main developer *June 2019 – January 2021*
  - *Developed a mobile application in Swift for improving users' health. Convinced early investor, a professional in health insurance.*

OUTREACH AND SERVICE

- **Volunteering for Chromosomic Disorders**
  I am particularly concerned about chromosomal disorders. I have participated in multiple events to raise awareness, collect money for research, and better integration of these individuals into society.
  - **Regatta Carla for the RETT syndrome 2018**
    Helped with the social media and promotion of a sailing regatta to raise funds for research on the RETT syndrome.

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- **Nastic Genuine 2018**
  Volunteered in the organization of a Spanish National football tournament for people with Down syndrome.
- **La Marato de TV3 2012-2018**
  On multiple occasions, I performed in music, both playing the baroque guitar and singing, to collect money for multiple causes, one of which was collecting money towards research on chromosomic disorders.

- **Mentorship for students 2018 - Present**
  Other things I care about is helping students from Spain achieve their maximal academic potential, mentor students to get into top programs in Europe and the US, and mentor them on career path.
  - **Google Developer Student Club of Lausanne November 2022**
    Gave a talk explaining my experience during my internship at Microsoft and the whole process of job finding in the United States.

SPOTLIGHT TALKS

- **Interview for a technology program in RTVE**
  I shared my research at MIT to non-specialized audiences by interviewing for the largest television broadcaster in Spain, RTVE. I will have a 2 min appearance. The episode has not been released yet but should be released in the beginning of 2023.
- **Research Presentation**
  I have presented my research in Reinforcement Learning and Robotics in multiple universities: MIT, Harvard University and University of Washington
- **Google Developer Student Club of Lausanne**
  Apart from mentoring students on a one-to-one basis I also participate to bigger reach events such as for the Google Developer Student Club.

OTHER ACTIVITIES

- **Running** Cambridge, MA
  2022 - Present
  I started running seriously very recently, and it has become a big part of my weekly routine to train my mind and body.
  - **Nice Ironman 70.3 2023** *26th June 2023*
    I ran a half-ironman in the city of Nice. This is one of the toughest triathlon races there are, consisting in a total of 70.3 miles between swimming, biking and running, and I completed it in under 6hours 43 minutes.
  - **Boston Half Marathon 2022** *13th November 2022*
    The BBA, Boston Half Marathon was my first running race ever. I am proud of my performance having ran it in 1 hour 42 minutes.

- **Sailing** Salou, Spain
  2008 - Present
  - **Windsurf World Championship 2017 & 2018**
    I competed at the highest level in windsurfing for three years. I was sailing in the Raceboard modality; this is a longer table for upwind and downwind sailing.
  - **29er National Championship 2015 & 2016**
    I competed at international level in 29er sailboat. This is a two-handed sailboat, considered one of the toughest boats to sail because of its instability.
  - **Optimist Catalan Championship 2010 – 2014**

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I competed at regional level in the optimist sailboat. This is a single-handed sailboat for kids up to the age of 16. This modality gets very competitive, and I participated at the international regatta of Riva del Garda, Italy, with up more than 1000 participants.

- **Undergraduate Music degree** Vila-seca, Spain
  Conservatory of music *2008 - 2018*
  Music used to be a very big part of my life, I did 10 years of music degree, which used to take from three to five afternoons per week depending on the semester. I specialized in Baroque Guitar and the Lute and learned from an internationally known musician in the field, William Waters. I played in Chamber Music accompanied by both voice and other string instruments, guitars, and violins.

SKILLS

- **Languages**
  - **Catalan -** *mother tongue*
  - **Spanish -** *mother tongue.*
  - **French -** *professional proficiency*

- **Programming Languages**
  - *Python, Javascript, C, C++, Scala, Spark, Kotlin SQL, HTML, CUDA, VHDL, MIPS assembly*
  - *ML libraries: Pytorch, JAX, TensorFlow, Pytorch Lightning, Weights and Biases*
  - *Robotics Simulators: Mujoco, Pybullet, Drake*